TECH WITH SHIVANSH

STATISTICS FOR DATA ANALYTICS & DATA SCIENCE

COMPLETE TOPICS LIST (PAGE 1 OF 2)

1 STATISTICS FUNDAMENTALS

- What is Statistics?
- Importance of Statistics in Data Analytics
- Applications in Real Life Business
- Population vs Sample
- Parameter vs Statistic
- Census vs Sampling
- Variables and Observations
- Types of Data
- Qualitative vs Quantitative Data
- Discrete vs Continuous Data
- Levels of Measurement
- Primary vs Secondary Data
- Data Collection Methods
- Sampling Basics & Sampling Techniques
- Sampling Bias & Sampling Error

2 DESCRIPTIVE STATISTICS

- Frequency Distribution
- Frequency Table
- Mean (Arithmetic Mean)
- Weighted Mean
- Median
- Mode
- Geometric Mean
- Harmonic Mean
- Range
- Variance
- Standard Deviation
- Mean Absolute Deviation (MAD)
- Coefficient of Variation (CV)
- Percentiles
- Quartiles
- Interquartile Range (IQR)
- Five Number Summary
- Measures of Central Tendency
- Measures of Dispersion

3 DATA DISTRIBUTION & VISUALIZATION

- Distribution of Data
- Symmetric vs Asymmetric Distribution
- Skewness: Positive, Negative & Zero
- Kurtosis (Introduction)
- Outliers: Detection & Treatment
- Box Plot (Box and Whisker Plot)
- Histogram
- Frequency Polygon & Ogive
- Stem and Leaf Plot
- Normal Distribution
- Empirical Rule (68–95–99.7 Rule)
- Z-Score
- Standardization of Data

4 PROBABILITY

- Introduction to Probability
- Experiment, Trial & Outcome
- Sample Space
- Events
- Types of Events
- Basic Probability
- Complementary Events
- Addition Rule of Probability
- Multiplication Rule of Probability
- Conditional Probability
- Independent Events
- Dependent Events
- Law of Total Probability
- Bayes' Theorem
- Applications of Probability in Business
- Real Life Probability Examples

5 RANDOM VARIABLES & PROBABILITY DISTRIBUTIONS

- Random Variable: Definition
- Discrete Random Variables
- Continuous Random Variables
- Probability Mass Function (PMF)
- Probability Density Function (PDF)
- Expected Value (Mean of Random Variable)
- Variance & Standard Deviation of Random Variable
- Binomial Distribution
- Poisson Distribution
- Normal Distribution in Depth
- Standard Normal Distribution
- Properties of Normal Distribution
- Sampling Distribution (Introduction)

6 SAMPLING DISTRIBUTION & CENTRAL LIMIT THEOREM

- Sampling Distribution of the Mean
- Sampling Distribution of Proportions
- Central Limit Theorem (CLT)
- Conditions for CLT
- Standard Error
- Effect of Sample Size
- Point Estimate vs Interval Estimate
- Bias and Unbiased Estimators

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BUILD IN-DEMAND SKILLS | SOLVE REAL BUSINESS PROBLEMS | MAKE BETTER DATA-DRIVEN DECISIONS | BOOST YOUR ANALYTICS CAREER

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STATISTICS FOR DATA ANALYTICS & DATA SCIENCE

COMPLETE TOPICS LIST (PAGE 2 OF 2)

7 ESTIMATION & CONFIDENCE INTERVALS

- Point Estimation vs Interval Estimation
- Confidence Interval: Definition
- Confidence Level
- Margin of Error
- Confidence Interval for Mean (σ Known)
- Confidence Interval for Mean (σ Unknown)
- Confidence Interval for Proportion
- Interpretation of Confidence Interval
- Common Mistakes in Confidence Intervals
- Use Cases in Business & Analytics

8 HYPOTHESIS TESTING BASICS

- Introduction to Hypothesis Testing
- Null Hypothesis (H_0)
- Alternative Hypothesis (H_1)
- One-tailed vs Two-tailed Test
- Level of Significance (α)
- Test Statistic
- P-value
- Critical Value Approach
- Decision Rule
- Type I Error (False Positive)
- Type II Error (False Negative)
- Power of a Test
- Statistical Significance vs Practical Significance
- Steps in Hypothesis Testing
- Business Interpretation of Hypothesis Testing

9 STATISTICAL TESTS FOR MEANS

- Z-Test: When and Why?
- One Sample Z-Test
- Two Sample Z-Test
- Assumptions of Z-Test
- T-Test: When and Why?
- One Sample T-Test
- Independent Two Sample T-Test
- Paired Sample T-Test
- Assumptions of T-Test
- Choosing Between Z-Test and T-Test
- Excel Implementation of Z & T Tests
- Business Use Cases

10 CHI-SQUARE TESTS

- Chi-Square Test: Overview
- Chi-Square Goodness of Fit Test
- Chi-Square Test of Independence
- Expected Frequency
- Assumptions
- Steps in Chi-Square Test
- Excel Implementation
- Real Life Business Applications

11 ANOVA (ANALYSIS OF VARIANCE)

- What is ANOVA?
- When to Use ANOVA?
- One Way ANOVA
- ANOVA Table (SS, df, MS, F-value, p-value)
- Assumptions of ANOVA
- Post Hoc Tests (Tukey, Bonferroni – Concept)
- Excel Implementation
- Business Applications of ANOVA

12 CORRELATION & REGRESSION ANALYSIS

- Covariance: Definition
- Correlation: Definition
- Pearson Correlation Coefficient (r)
- Range and Interpretation of r
- Correlation vs Causation
- Scatter Plot
- Simple Linear Regression
- Regression Equation (Y = a + bX)
- Slope (b) & Intercept (a)
- R-squared (R^2) Meaning
- Adjusted R-squared
- Residuals: Concept
- Assumptions of Linear Regression
- Excel Regression
- Business Case Studies

13 ADVANCED TOPICS & APPLICATIONS

- Multiple Linear Regression (Conceptual)
- Multicollinearity
- Heteroscedasticity
- Autocorrelation
- Outliers & Leverage Points
- Effect Size
- Confidence Interval for Regression Coefficients
- A/B Testing: Introduction
- A/B Testing: Design Principles
- Statistical Testing for A/B Tests
- Experimental Design Basics
- Seasonality & Time Series Basics (Intro)
- Choosing the Right Statistical Technique
- Common Statistical Pitfalls
- Statistics for Machine Learning (Conceptual)
- Business Dashboard Metrics & KPIs using Statistics

WHAT YOU WILL ACHIEVE

- Strong foundation in Statistics
- Become confident in Data Analytics & Data Science roles
- Apply Statistics to real business problems
- Crack interviews with conceptual clarity

EXCEL SKILLS YOU WILL MASTER

- Statistical Functions
- Data Analysis ToolPak
- Data Visualization
- Hypothesis Testing
- Correlation & Regression
- ANOVA & T-Tests in Excel
- Automated Statistical Reports

LEARNING APPROACH

- Concept → Example
- Formula → Calculation
- Excel Practice
- Business Application
- Interview Preparation

WHO SHOULD LEARN?

- Data Analyst Aspirants
- Data Scientist Aspirants
- Business Analysts
- MBA Students
- Finance & Marketing Professionals
- Anyone who wants to learn Statistics from scratch

UNDERSTAND STATISTICS. APPLY IT. SOLVE REAL BUSINESS PROBLEMS.

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